

| Paper No. | paper_id | citation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | doi_or_url                                                                                                        | sample_size                                                                 | n_events                            | prevalence | n_classifiers                             | winner                          | winner_score                  | runner_up_score                                                                                    | protocol                                                                                                             | test_reported                                                                                                                          | claim_in_abstract                                                                 | notes                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| 1         |          | Dominguez-Rodriguez S, Serna-Pascual M, Dietto A, Barnabas S, Zuidewind P, Dobbels E, Danaviah S, Behuhuma O, Lain MG, Vaz P, Fernández-Luis S, Nhampossa T, Lopez-Varela E, Chwombi K, Liberty A, Volterri A, Maiga AI, Rossi P, Glaquinto C, Kuhn L, Rojo P, Tagarro A. EPICAL Consortium. Machine learning outperformed logistic regression classification even with limit sample size: A model to predict pediatric HIV mortality and clinical progression to AIDS. <i>PLoS One</i> . 2022 Oct 14;17(10):e0276116. doi: 10.1371/journal.pone.0276116. PMID: 36240212; PMCID: PMC9565414. | <a href="https://pubmed.ncbi.nlm.nih.gov/36240212/">https://pubmed.ncbi.nlm.nih.gov/36240212/</a>                 | 100                                                                         | 33                                  | 33.00%     | 7                                         | Random Forest<br>Random Forest  | AUC = 0.732                   | AUC = 0.696 (KNN)                                                                                  | 70/30 train-validation split + repeated 10-fold CV for tuning                                                        | yes                                                                                                                                    | yes                                                                               | Small-sample study (n=100); 7 classifiers compared. RF was the best model (AUC 0.732), followed by KNN (0.696). Repeated 10-fold CV used for tuning; performance comparisons reported.                                                                                                                                                                                                                                            |
| 2         |          | Góngora Alonso S, Marques G, Agarwal D, De la Torre Díez I, Franco-Martin M. Comparison of Machine Learning Algorithms in the Prediction of Hospitalized Patients with Schizophrenia. <i>Sensors</i> . 2022.                                                                                                                                                                                                                                                                                                                                                                                 | <a href="https://pmc.ncbi.nlm.nih.gov/articles/PMC9003328/">https://pmc.ncbi.nlm.nih.gov/articles/PMC9003328/</a> | 11,884                                                                      | 5,968                               | 50.20%     | 6                                         |                                 | AUC = 0.796                   | AUC = 0.765                                                                                        | 10-fold stratified cross-validation                                                                                  | Yes                                                                                                                                    | Yes                                                                               | Applied clinical ML study using structured/tabular hospital admission records. Six classifiers (Random Forest, k-NN, AdaBoost, Naive Bayes, Decision Tree, and SVM) were compared on the same dataset. The authors explicitly identify Random Forest as the best-performing model. The paper reports 11,884 admission records from 6,933 patients; 5,968 records correspond to schizophrenia and 5,916 to other mental disorders. |
| 3         |          | Yihan Zhang, Dong Yang, Zifeng Liu, Chaojin Chen, Mian Ge, Xiang Li, Tongsen Luo, Zhengdong Wu, Chenguang Shi, Bohan Wang, Xiaoshuai Huang, Xiaodong Zhang, Shaoli Zhou, Ziqing Hei (2021). "An explainable supervised machine learning predictor of acute kidney injury after adult deceased donor liver transplantation". <i>Journal of Translational Medicine</i>                                                                                                                                                                                                                         | <a href="https://pubmed.ncbi.nlm.nih.gov/34321016/">https://pubmed.ncbi.nlm.nih.gov/34321016/</a>                 | 780 included cases for the internal analysis; 894 cases initially collected | 430 AKI cases                       | 55.10%     | 5 — LR, SVM, RF, GBM, AdaBoost            | Gradient Boosting Machine (GBM) | AUC = 0.76 (95% CI 0.70–0.82) | Not clearly reported in the abstract/text as a single second-best AUC                              | 70/30 development/internal-validation split + 5-fold CV for hyperparameter tuning + external validation on 195 cases | Yes — 95% CIs reported for performance; model comparisons were evaluated, although not a simple significance test between every pair   | Yes — GBM is presented as the best-performing model                               | Important distinction: 894 cases were collected, but 780 were included in the internal analysis. There was also an independent external dataset of 195 patients. Use 780 as the model-analysis sample size and mention 894 in notes.                                                                                                                                                                                              |
| 4         |          | Munetoshi Akazawa, Kazunori Hashimoto, Katsuhiko Noda, Kaname Yoshida (2021). "The application of machine learning for predicting recurrence in patients with early-stage endometrial cancer: a pilot study". <i>Obstetrics &amp; Gynecology Science</i>                                                                                                                                                                                                                                                                                                                                     | <a href="https://pubmed.ncbi.nlm.nih.gov/33371658/">https://pubmed.ncbi.nlm.nih.gov/33371658/</a>                 | 75 patients                                                                 | 12 recurrence cases                 | 16.00%     | 5 — SVM, RF, DT, LR, Boosted Tree         | Logistic Regression (LR)        | AUC = 0.53                    | Boosted Tree: AUC = 0.52                                                                           | 80/20 random train-test split + 5-fold cross-validation                                                              | No — no significance test/CI specifically comparing classifier performance                                                             | Yes — abstract states LR was the best predictive model                            | Very small dataset. Interestingly, SVM had the highest accuracy (0.82), but LR had the highest AUC (0.53), so LR was declared the best model.                                                                                                                                                                                                                                                                                     |
| 5         |          | Jihu Wei, Shijin Lu, Wencai Liu, He Liu, Lin Feng, Yizi Tao, Zhanjin Pu, Qiang Liu, Zhaochi Hu, Haosheng Wang, Wenle Li, Wei Kang, Chengliang Yin, Zhe Feng (2023). "A machine learning-based model for clinical prediction of distal metastasis in chondrosarcoma: a multicenter, retrospective study". <i>PeerJ</i>                                                                                                                                                                                                                                                                        | <a href="https://pubmed.ncbi.nlm.nih.gov/38130920/">https://pubmed.ncbi.nlm.nih.gov/38130920/</a>                 | 1,385 patients                                                              | 82 patients with distant metastasis | 5.90%      | 6 — LR, Naive Bayes, DT, RF, GBM, XGBoost | XGBoost (XGB)                   | AUC = 0.985                   | Reported range for all models = 0.911–0.985; exact runner-up AUC is not clearly stated in the text | 10-fold cross-validation + external validation                                                                       | Yes — paper states XGBoost significantly outperformed the other ML models                                                              | Yes — abstract explicitly identifies XGBoost as having the highest AUC/best model | Strong fit overall. Uses readily available clinical variables rather than imaging/radiomics. Training data: 1,281 SEER patients; external validation: 104 patients from 3 Chinese medical centers. However, the exact runner-up AUC is not clearly given in the text, so mark it as NR rather than guessing.                                                                                                                      |
| 6         |          | Zhong Z, Yuan X, Liu S, Yang Y, Liu F. Machine learning prediction models for prognosis of critically ill patients after open-heart surgery. <i>Scientific Reports</i> . 2021;11:3384.                                                                                                                                                                                                                                                                                                                                                                                                       | <a href="https://pubmed.ncbi.nlm.nih.gov/33564090/">https://pubmed.ncbi.nlm.nih.gov/33564090/</a>                 | 6844                                                                        | 215                                 | 3.10%      | 4                                         | XGBoost                         | 0.90 AUC                      | 0.88 AUC (Random Forest)                                                                           | 80/20 train-test split; 5-fold cross-validation + grid search on training data                                       | No CI or statistical significance test for model-performance comparison reported; chi-square/Wilcoxon tests used for other comparisons | Yes                                                                               | MIMIC-III; tabular clinical data; outcome used here = 30-day mortality; 4 classifiers: XGBoost, RF, LR, ANN; XGBoost selected as final/best model.                                                                                                                                                                                                                                                                                |
| 7         |          | Meng L, Wei T, Fan R, Su H, Liu J, Wang L, Huang X, Qi Y, Li X. Development and validation of a machine learning model to predict venous thromboembolism among hospitalized cancer patients. <i>Asia-Pacific Journal of Oncology Nursing</i> . 2022;9(12):100128.                                                                                                                                                                                                                                                                                                                            | <a href="https://pubmed.ncbi.nlm.nih.gov/36276886/">https://pubmed.ncbi.nlm.nih.gov/36276886/</a>                 | 1100                                                                        | 340                                 | 30.9       | 4                                         | XGBoost                         | 0.818 AUROC                   | 0.759 AUROC (SVM)                                                                                  | Random 80/20 split; 80% training, 20% testing                                                                        | NR in abstract; verify full-text statistical comparison                                                                                | YES                                                                               | Retrospective case-control study; hospitalized cancer patients; clinical, tumor, treatment and laboratory variables; classifiers were Logistic Regression, SVM, Random Forest and XGBoost. XGBoost explicitly identified as optimal.                                                                                                                                                                                              |
| 8         |          | Qu C, Gao L, Yu XQ, Wei M, Fang GQ, He J, Cao LX, Ke L, Tong ZH, Li WQ. Machine Learning Models of Acute Kidney Injury Prediction in Acute Pancreatitis Patients. <i>Gastroenterology Research and Practice</i> . 2020;2020:3431290.                                                                                                                                                                                                                                                                                                                                                         | <a href="https://pubmed.ncbi.nlm.nih.gov/33061958/">https://pubmed.ncbi.nlm.nih.gov/33061958/</a>                 | 334                                                                         | 80                                  | 23.95      | 5                                         | XGBoost                         | 0.9193 AUC                    | 0.8821 AUC (Random Forest)                                                                         | 70/30 train-test split; 10-fold cross-validation for parameter tuning on training data                               | No — no CI/significance test reported for model-performance comparison                                                                 | YES                                                                               | 334 acute pancreatitis patients; 80 developed AKI and 254 did not. Five models: LR, CART, XGBoost, SVM, RF. XGBoost explicitly reported as best. 23 clinical/demographic/laboratory features.                                                                                                                                                                                                                                     |

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|    |  | Qian Q, Wu J, Wang J, Sun H, Yang L. Prediction Models for AKI in ICU: A Comparative Study. Int J Gen Med. 2021 Feb;25(14):623-632. doi: 10.2147/IJGM.S289671. PMID: 33664585; PMCID: PMC7921622.                                                                                                                               | <a href="https://pubmed.ncbi.nlm.nih.gov/33664585/">https://pubmed.ncbi.nlm.nih.gov/33664585/</a>                             | 17205          | 3,929 AKI cases | 22.84% (3,929/17,205) | 6                                                                                                                               | LightGBM            | 0.905 AUC                         | XGBoost AUC = 0.760                              | Random 80/20 train-test split; class balancing repeated 10 times for each model; no cross-validation                                                      | Yes — statistical comparison—P-values reported for AUC differences                      | Yes                                                               | MIMIC-III ICU data; 17,205 patients; 17 predictor variables; models : LR, SVM, RF, XGBoost, LightGBM and CNN. LightGBM reported as best; XGBoost second.                                                                                                                                                                                                                                                                                                                                                |
| 9  |  | Soleimani H, Najdaghi S, Narimani Davani D, Dasjerdi P, Samimisadeh P, Shayesteh H, Sattariabadi B, Masoudkabr F, Ashraf H, Mehrani M, Jenab Y, Hosseini K. Predicting In-Hospital Mortality in Patients With Acute Myocardial Infarction: A Comparison of Machine Learning Approaches. Clinical Cardiology. 2025;48(4):e70124. | <a href="https://doi.org/10.1002/clc.70124">https://doi.org/10.1002/clc.70124</a>                                             | 7422           | 129             | 1.74                  | 7                                                                                                                               | Random Forest       | 0.924 AUC                         | 0.905 AUC (XGBoost)                              | 70% training / 30% testing; 5-fold cross-validation; SMOTE for class imbalance                                                                            | Yes — 95% CI for RF AUC: 0.893–0.954                                                    | Yes                                                               | 7422 AMI patients treated with PCI; 58 clinical, demographic and laboratory variables; binary outcome = in-hospital mortality. Seven ML algorithms compared. RF was the best-performing model.                                                                                                                                                                                                                                                                                                          |
| 10 |  | Akazawa M, Hashimoto K, Noda K, Yoshida K. Machine learning approach for the prediction of postpartum hemorrhage in vaginal birth. Scientific Reports. 2021.                                                                                                                                                                    | <a href="https://pmc.ncbi.nlm.nih.gov/articles/PMC8604915/#Abst1">https://pmc.ncbi.nlm.nih.gov/articles/PMC8604915/#Abst1</a> | 9894           | 188             | 1.90%                 | 5                                                                                                                               | Logistic Regression | AUC = 0.708                       | AUC = 0.651                                      | 80/20 train-test split; internal validation/cross-validation; repeated 5 times                                                                            | Yes — comparative claim; the abstract reports the “best learning model” with AUC 0.708. | Yes                                                               | Five ML classifiers: Logistic Regression, SVM, Random Forest, Boosting Trees (XGBoost), Decision Tree. Deep learning was also evaluated separately. Logistic Regression had the best AUC (0.708), followed by Random Forest (0.651), Boosted Trees (0.634), and Decision Tree (0.596). The paper's Table 3 does not show the SVM AUC in the extracted table, so do not infer it. PPH defined as blood loss >1000 mL.                                                                                    |
| 11 |  | Sayed AFA, Arafa MA, El-Ninr NA, Banawan KAS, Abdou MS. Comparison of machine learning classification and regression models for prediction of academic performance among postgraduate public health students. Scientific Reports. 2025.                                                                                         | <a href="https://pmc.ncbi.nlm.nih.gov/articles/PMC12715227/">https://pmc.ncbi.nlm.nih.gov/articles/PMC12715227/</a>           | 922            | 113             | 12.30%                | 9                                                                                                                               | Random Forest       | AUC = 0.87                        | AUC = 0.86                                       | 75/25 train-test split with 5-fold cross-validation                                                                                                       | Yes                                                                                     | Yes                                                               | Nine classification models were compared. Random Forest was reported as the best classifier with accuracy = 71.43%, precision = 0.81, recall = 0.53, AUC = 0.87, F1 = 0.68, and MCC = 0.52. Ensemble, kNN, Logistic Regression and RBF had AUC = 0.86 where reported. The lowest-performing class (less than expected/un satisfactory) represented 12.3% of 922 students, approximately 113 students. The paper also compares regression models, but those are separate from the classifier comparison. |
| 12 |  | Mao J, Chao K, Jiang FL, et al. Comparison and development of machine learning for thalidomide-induced peripheral neuropathy prediction of refractory Crohn's disease in Chinese population. World Journal of Gastroenterology. 2023.                                                                                           | <a href="https://pmc.ncbi.nlm.nih.gov/articles/PMC10324537/">https://pmc.ncbi.nlm.nih.gov/articles/PMC10324537/</a>           | 164            | 59              | 36%                   | 5                                                                                                                               | XGBoost             | Accuracy = 0.818                  |                                                  | 80/20 random train-test split; SMOTE applied to training and test sets; 5-fold cross-validation; hyperparameters tested across 1000 random dataset splits | Yes                                                                                     | Yes                                                               | Five models compared: GBOT, Extremely Randomized Trees, Random Forest, Logistic Regression and XGBoost. XGBoost reported as the best-performing model.                                                                                                                                                                                                                                                                                                                                                  |
| 13 |  | Dengler NF, Madai VI, Unteroberdorster M, Zihni E, Brune SC, Hilbert A, et al. Outcome prediction in aneurysmal subarachnoid hemorrhage: a comparison of machine learning methods and established clinico-radiological scores. Neurosurgical Review. 2021.                                                                      | <a href="https://pmc.ncbi.nlm.nih.gov/articles/PMC10324537/">https://pmc.ncbi.nlm.nih.gov/articles/PMC10324537/</a>           | 388            | 179             | 46.10%                | 7                                                                                                                               | Tree Boosting       | AUC = 0.78                        | AUC = 0.77                                       |                                                                                                                                                           | Yes                                                                                     | Yes                                                               | INCLUDE. Applied clinical ML study using clinical/radiographic tabular data. Seven ML algorithms were compared on the same cohort. Tree Boosting achieved the highest reported AUC (0.78); the other models were around 0.77, with Naive Bayes 0.75. The paper uses a somewhat hedged comparative conclusion (“no relevant difference”), but it explicitly identifies the highest-performing model, so it satisfies criterion #4.                                                                       |
| 14 |  | Dong Y, Wen H, Lu C, Li J, Zheng Q. Predicting depression risk with machine learning models: identifying familial, personal, and dietary determinants. BMC Psychiatry. 2025.                                                                                                                                                    | <a href="https://pubmed.ncbi.nlm.nih.gov/33474607/">https://pubmed.ncbi.nlm.nih.gov/33474607/</a>                             | 7108           | 2574            | 36.21%                | 11                                                                                                                              | Random Forest       | AUC = 0.705                       | AUC = 0.719                                      | Random 4:1 train/test split; 10-fold cross-validation; 50 shuffles.                                                                                       | Yes                                                                                     | Yes                                                               | Winner definition is ambiguous. Abstract identifies RF as superior, but test-set AUC is highest for LR and Lasso (0.719), while RF = 0.705.                                                                                                                                                                                                                                                                                                                                                             |
| 15 |  | Jeon Y, Kim YJ, Jeon J, et al.                                                                                                                                                                                                                                                                                                  | <a href="https://pmc.ncbi.nlm.nih.gov/articles/PMC12487390/">https://pmc.ncbi.nlm.nih.gov/articles/PMC12487390/</a>           | 961            | 127             | 13.20%                | 4 - Logistic Regression                                                                                                         | Support Vector      | 0.831 AUC                         | 0.826 AUC                                        | Random 80/20                                                                                                                                              | Yes —                                                                                   | Yes                                                               | Four classifiers compared: LR, SVM, RF and XGBoost. SVM                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 16 |  | Li X, Zhu Y, Zhao W, Shi                                                                                                                                                                                                                                                                                                        | <a href="https://pmc.ncbi.nlm.nih.gov/articles/PMC10723658/">https://pmc.ncbi.nlm.nih.gov/articles/PMC10723658/</a>           | 8527           | 1406            | 16.50%                | 6 - Logistic Regression                                                                                                         | XGBoost             | 0.86                              | 0.841                                            | Hierarchical 5-fold                                                                                                                                       | Yes — testing                                                                           | Yes                                                               | Six ML models compared. XGBoost had the highest AUC                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 17 |  | Xi Y, Wang H, Sun N,                                                                                                                                                                                                                                                                                                            | <a href="https://pmc.ncbi.nlm.nih.gov/articles/PMC9701715/">https://pmc.ncbi.nlm.nih.gov/articles/PMC9701715/</a>             | 143,043        | 30,903          | 21.60%                | 7- LR with Elastic-Net                                                                                                          | XGBoost Ensemble    | 0.760 AUROC                       | 0.754 AUROC                                      | 80/20                                                                                                                                                     | Yes —20%                                                                                | Yes                                                               | Six ML/ml/modeling methods plus benchmark LR were                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 18 |  | Park SW, Yeo NY, Kang S, et al. (2024). Early Prediction of Mortality for Septic Patients Visiting Emergency Room Based on Explainable Machine Learning.                                                                                                                                                                        | <a href="https://pmc.ncbi.nlm.nih.gov/articles/PMC10843974/">https://pmc.ncbi.nlm.nih.gov/articles/PMC10843974/</a>           | 5,112          | 1,455           | 28.50%                | 6 - Logistic Regression (LR)<br>Support Vector Machine (SVM)<br>Random Forest (RF)<br>XGBoost                                   | CatBoost            | 0.800 AUC                         | 0.797 AUC                                        | 80/20 train-test split, stratified 5-fold cross-validation during training; repeated random sampling                                                      | Yes — separate 20% test dataset                                                         | Yes                                                               | Six classifiers compared using 44 clinical variables. CatBoost had the highest AUC (0.800), followed by XGBoost (0.797). The paper also evaluates the six models using SOFA components, where XGBoost was best (AUC 0.678). For the primary clinical-variable comparison, use CatBoost as the winner.                                                                                                                                                                                                   |
| 19 |  | Sung Hyun Yoon, Cheolho Kwon, Yeongho Choi, Hyung-Jun Kim, Jilhang Kim, Young Hoan Kim (2026). "Machine learning improves prediction of pulmonary thromboembolism and reduces unnecessary computed tomography scans in the emergency department". Scientific Reports                                                            | <a href="https://pubmed.ncbi.nlm.nih.gov/41513701/">https://pubmed.ncbi.nlm.nih.gov/41513701/</a>                             | 2,525 patients | 573 PTE cases   | 22.70%                | 7 — boosted trees/XGBoost, Random Forest, Logistic Regression, Elastic Net, linear SVM, radial SVM, feed-forward neural network | XGBoost             | AUC = 0.814 (95% CI: 0.759–0.862) | Random Forest: AUC = 0.801 (95% CI: 0.753–0.850) | 80/20 train-test split + 5-fold cross-validation for hyperparameter tuning + bootstrap (1,000 resamples) for 95% Cis                                      | Yes — 95% CIs reported; p-values also reported for CTPA reduction                       | Yes — abstract identifies XGBoost as the highest-performing model | Six classifiers compared using 44 clinical variables. CatBoost had the highest AUC (0.800), followed by XGBoost (0.797). The paper also evaluates the six models using SOFA components, where XGBoost was best (AUC 0.678). For the primary clinical-variable comparison, use CatBoost as the winner.                                                                                                                                                                                                   |
| 20 |  |                                                                                                                                                                                                                                                                                                                                 | <a href="https://pubmed.ncbi.nlm.nih.gov/41513701/">https://pubmed.ncbi.nlm.nih.gov/41513701/</a>                             | 2,525 patients | 573 PTE cases   | 22.70%                |                                                                                                                                 |                     |                                   |                                                  |                                                                                                                                                           |                                                                                         |                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |